Discrete Mathematics

Tut-6 (Solution)

Q1

a) No Euler path
No Euler Circuit

b) No Euler Circuit
Euler path Exists

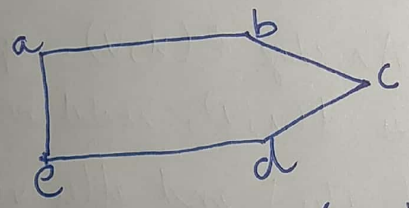
c) Euler Circuit : a e a e c b e d c d b a

d) No Euler Circuit
No Euler Path

e) Euler Path Exists
No Euler Circuit

Q2

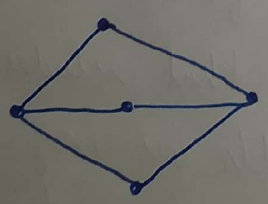
a) There exists a Hamiltonian Circuit (a b c d e a)



b) There is no Hamiltonian Circuit.

Q3

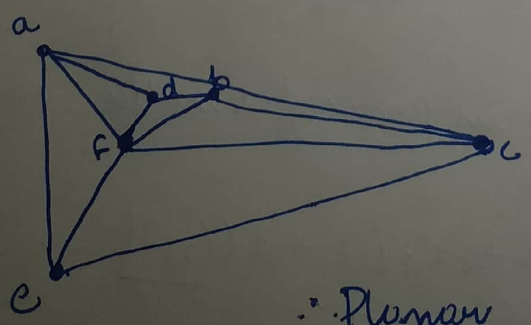
a)



Planar

b) Not Planar

c)



∴ Planar

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4) The Travelling Salesman Problem consists of a salesman and a set of cities. The salesman has to visit each one of the cities starting from a certain one and returning to the same city. The challenge of the problem is that the travelling salesman wants to minimize the total length of the trip.

Travelling Salesman Problem can be described as follows:

$TSP = \{(G, f, t) : G = (V, E)$ a complete graph,

f is a function $V \times V \to Z$

$t \in Z$

G is a graph that contains a travelling salesman tour with cost that doesn't exceed 't' $\}$

<u>Process to solve:</u>

Let us consider a graph $G = (V, E)$ where V is set of cities & E is set of weight edge. An edge $e(u,v)$, represents u & v are connected. Dist b/w u & v is $d(u,v)$, should be non-negative

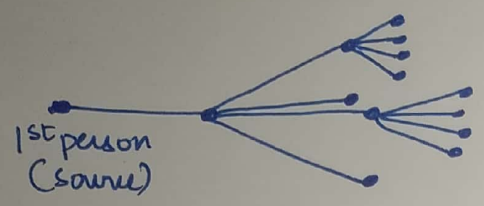
Suppose we have started at city 1 and after visiting some cities now we are in city j. Hence this is a partial tour.

We have to know j, as this will determine which cities to

know what all cities are visited

5) An empty graph (no edges)

6)



We assume that 1st person sends the letter to one person, then we need to find all the nodes with degree 5 in order to find no. of people who sent out letters not including the who started the chain.

Using handshake theorem,

$$5x + 101 = 2(4x+1)$$
$$5x + 101 = 8x + 2$$
$$3x = 99$$
$$x = 33$$

No. of people who sent letter $= x+1 = \underline{\underline{34}}$

7) No. of edges in a tree $= n - 1$
$= 10000 - 1$
$= \underline{\underline{9999}}$

8) a) 2

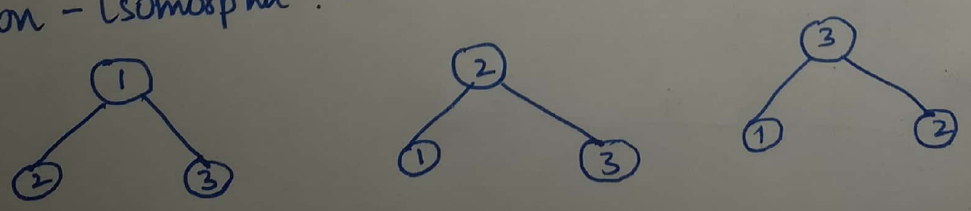
(i) $H_3C - CH_2 - CH_2 - CH_3$

(ii) $H_3C - \underset{\underset{CH_3}{|}}{CH} - CH_3$

b) 1

(i) $H_3C - CH_2 - CH_3$

9) Non-isomorphic : 3



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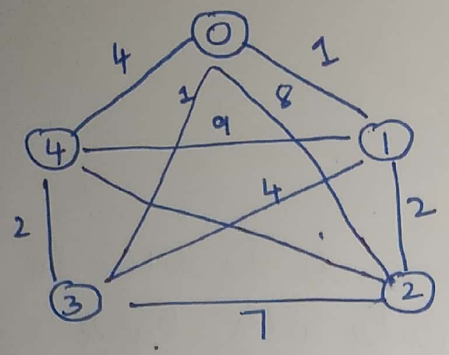
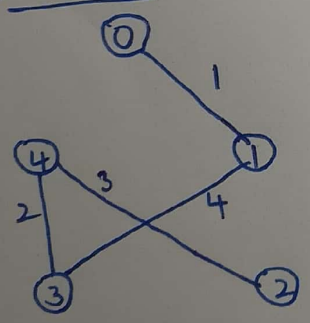


As 0 need to be leaf, only one edge can exist.

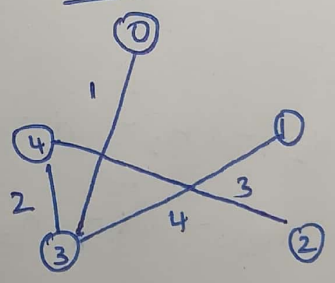
Kruskal

Edges	Weight (Sorted)
0-1	1
0-3	1
3-4	2
2-4	3
1-3	4
0-4	4
2-3	7
0-2	8
1-4	9
1-2	12

Possibility 1



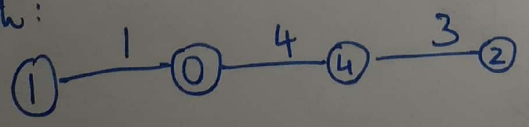
Possibility 2



Minimum wt in both the cases is 10.

b) Min-weight = 8

Path:



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